How Do Language Models Represent Policy Text? A Proxy Grounded Mechanistic Analysis of Policy Features

Seunghyun Yoo
Purdue University
yoo236@purdue.edu

Abstract

We study how large language models internally represent and process policy text. Our starting point is AGORA, a living collection of AI governance documents with segment level text, metadata, quality flags, and structured tag annotations (Arnold et al., 2024). Rather than collapsing policy passages into a single ideological axis, we treat AGORA policy tags as observable proxies for policy relevant concepts and ask whether internal features discovered from one proxy generalize to related proxies, survive lexical stress, and causally influence held out policy judgments. The proposed methodology combines document grouped splits, metadata matched negatives, dense residual representations, sparse SAE features on Gemma 2 2B pretrained activations with aligned Gemma Scope dictionaries (Gemma Team, Google DeepMind, 2024; Lieberum et al., 2024), train only feature discovery, held out cross proxy transfer, keyword masking, family core feature consolidation, and intervention based causal testing. We additionally package this evaluation into a compact audit benchmark, but treat that benchmark as a supporting protocol rather than the paper's main identity. The current implementation already supports the full pipeline over 2,719 filtered AGORA segments and passes all benchmark preflight checks.

1 Introduction

Language models are increasingly used in workflows that touch governance and public policy, including retrieval, drafting, summarization, critique, and decision support. Yet we still know relatively little about how they internally process policy language. Most work either evaluates model outputs directly or studies internal computation on synthetic reasoning tasks rather than on naturally occurring policy text (Elhage et al., 2021; Conmy et al., 2023; Ameisen et al., 2025). That leaves an important gap: if language models are going to be used in governance settings, we need methods for asking what kinds of policy relevant structure they internally represent, how stable those representations are, and whether they actually matter for downstream judgments.

This paper takes that problem as its starting point. Our main research question is: *how do language models internally represent policy text and policy relevant features?* We focus on policy features rather than on abstract values because those features are directly observable in AGORA. AGORA provides segment level AI governance text together with existing tags over risks, harms, incentives, applications, and governance strategies (Arnold et al., 2024). These tags are not treated as a complete ontology. Instead, they serve as proxy anchors that let us discover and test internal representations in a grounded way.

From this main question follow three narrower ones. First, which policy features produce distinct and robust internal representations? Second, do related policy features share internal structure that generalizes across proxies? Third, are these discovered features causally involved in held out policy judgments? These questions are narrow enough to support tractable mechanistic analysis while still being broad enough to matter for technical governance.

Our answer is methodological rather than purely descriptive. We build an end to end pipeline that starts from AGORA policy segments, constructs matched negatives, extracts dense and sparse internal representations, performs train only proxy level feature discovery, evaluates held out transfer across related policy features, stress tests those features under keyword masking, and validates selected feature sets through intervention. We then package the evaluation component into a compact audit benchmark, but the benchmark is a wrapper around the methodology, not the main contribution itself.

This framing keeps the claim boundary clear. We do not begin by asking whether models directly encode public values. We first ask whether they contain structured internal policy feature representations that can be discovered, generalized, and causally tested. Only after those conditions are met does it become reasonable to discuss higher level interpretations that are *consistent with* broader policy concept families.

2 Related Work

Mechanistic interpretability studies neural models through internal algorithms, circuits, and representa-

tional structure (Elhage et al., 2021). Circuit discovery methods such as ACDC and attribution based tracing have made it increasingly feasible to localize task relevant computation in language models (Conmy et al., 2023; Ameisen et al., 2025). Sparse autoencoders and related dictionary learning methods are especially important because they decompose dense activations into sparse features that can be ranked, compared, and intervened on (Bricken et al., 2023; Cunningham et al., 2024; Marks et al., 2024). Gemma Scope extends this line of work to Gemma 2 by releasing aligned sparse autoencoders across layers and sites, making feature level analysis practical in a modern open model family (Lieberum et al., 2024). Transcoder style work further suggests that sparse representations can support tracing across layers rather than only within a single local feature set (Dunefsky et al., 2024).

Recent interpretability work also emphasizes that attractive examples are not enough. Stronger claims require held out validation, stress testing, and causal intervention (Arditi et al., 2024; Ameisen et al., 2025). We adopt that standard directly. In our setting, proxy aligned feature discovery is only the first step. A feature bank becomes interesting only when it transfers to related policy concepts, survives lexical masking, or produces selective behavioral changes under intervention.

On the data side, AGORA is well suited to this problem because it provides full text, summaries, metadata, and tag annotations for AI governance documents (Arnold et al., 2024). At the same time, AGORA tags are specific and context dependent. That makes them useful as observable policy proxies, but poor candidates for direct claims about abstract value representation. Our framework therefore treats AGORA tags as measurable anchors for internal analysis and reserves higher level policy interpretation for downstream discussion.

3 Policy Representation Setup

3.1 Policy Segments and Proxy Features

The unit of analysis is the AGORA segment. Each segment contains policy text together with quality flags such as whether the segment is operative, AI related, annotated, and validated, along with tag annotations for policy content (Arnold et al., 2024). In this project, a *policy feature* means a tagged policy relevant concept that can be operationalized from these existing AGORA annotations.

For the current core study, we focus on three related concept pairs:

1. **Bias** and **Discrimination**
2. **Privacy** and **Rights violation**
3. **Transparency** and **Interpretability**

These pairs were chosen because they let us study three things at once: proxy level separability, transfer across related concepts, and the possibility of shared internal structure. A single isolated proxy can show that a model separates one tag from a control set. A related pair lets us ask the stronger question of whether that separation reflects a reusable internal feature bank rather than a tag specific lexical shortcut.

3.2 Research Questions

The paper centers on four research questions.

1. **RQ1.** How do language models internally represent policy text and policy relevant features?
2. **RQ2.** Which policy features produce distinct and robust internal representations?
3. **RQ3.** Do related policy features share internal structure that generalizes across proxies?
4. **RQ4.** Are these discovered policy features causally involved in held out policy judgments?

These questions deliberately stay below the level of direct public value claims. Public values are relevant as a secondary lens for interpreting families of related policy features, but they are not the empirical target of the current analysis.

3.3 Claim Boundary

The methodology distinguishes three levels of claim.

1. **Proxy aligned feature claim.** The model contains internal features aligned with individual AGORA policy proxies.
2. **Shared policy feature claim.** Some discovered features generalize across related policy proxies.
3. **Higher level interpretation.** Such shared structure may be consistent with broader policy concept families.

The first two claims are direct empirical targets of this paper. The third is interpretive and should only be entertained when supported by transfer, robustness, and causal evidence.

4 Methodology

Figure 1 summarizes the full pipeline.

4.1 AGORA Segment Selection and Quality Filtering

We begin from AGORA segment level policy text and retain only examples that are AI related, operative, and annotated. Operationally, this means excluding segments marked as not AI related or non operative, and using annotated segments as the main analysis pool. Validated segments are not merged into the main pool. Instead, they define a higher confidence robustness subset for reruns and diagnostics.

This filtering yields 2,719 eligible segments in the current build. To prevent document specific leakage,

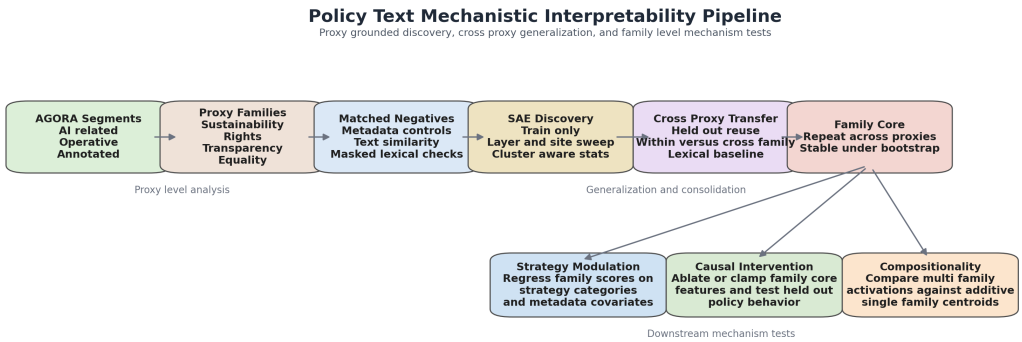

Figure 1: Proxy grounded methodology for policy representation analysis. Starting from AGORA policy segments, the pipeline filters high quality examples, constructs proxy specific manifests and matched negatives, extracts dense and sparse internal representations, performs train only feature discovery, evaluates held out transfer and lexical robustness, consolidates repeated sparse features into family core banks, and finally applies intervention based causal testing. The benchmark used later in the paper is a structured evaluation wrapper around these steps.

all train, development, and test splits are performed at the document level. Segments from the same source document therefore never appear on both sides of a held out evaluation.

4.2 Policy Proxy Definition

The next step is to convert AGORA annotations into concrete proxy tasks. We use existing AGORA tag annotations as observable policy features rather than inventing new labels. For the current core study, this produces six binary proxy tasks corresponding to the three concept pairs above. Each proxy becomes a positive set of policy segments to be contrasted with a carefully matched negative set.

This proxy grounded design matters because it makes the representational question testable. Instead of asking whether a model encodes an abstract concept directly, we ask whether it contains internal features aligned with specific policy proxies and whether those features generalize to nearby concepts.

4.3 Concept Pair Selection

The three selected concept pairs were chosen to balance interpretability, coverage, and transfer value. Bias and discrimination give a risk to harm contrast around equality relevant governance language. Privacy and rights violation provide a bridge between technical risk framing and legal or normative rights framing. Transparency and interpretability connect governance facing and technical model facing language. Together, these pairs let us test whether models store isolated proxy responses or broader reusable feature structure.

4.4 Matched Negative Construction

Proxy discovery is defined as *positive versus matched negative*, not positive versus the full remainder of the corpus. For each proxy and split, we build a negative set from examples that do not carry the target proxy but are close in text and metadata. Matching uses document metadata such as authority, jurisdiction, document form, year, and domain together with text similarity, so that the resulting contrast is not trivially solved by institution specific style or formal document structure alone.

This is a key design choice. Without matched negatives, a model could appear to represent a policy concept when it is really responding to shallow differences in source type, boilerplate, or topic cluster.

4.5 Internal Representation Extraction

We study two kinds of internal representations. The first is a dense baseline based on pooled residual activations. The second is a sparse representation obtained by encoding the same activations through aligned sparse autoencoders. The main model setting is Gemma 2 2B pretrained with Gemma Scope aligned sparse autoencoders on the residual stream (Gemma Team, Google DeepMind, 2024; Lieberum et al., 2024). We use `resid_post` as the main activation site and evaluate candidate layers $\{12, 16, 20, 24\}$. Mean pooling is used for the main runs and max pooling is reserved for robustness reruns.

Dense features are useful as a non sparse internal baseline. Sparse SAE features are useful because they support ranking, selection, reuse across tasks, and direct intervention.

4.6 Train Only Feature Discovery

For each proxy task, we compare sparse activations from the positive set and its matched negatives. Let $f(x) \in \mathbb{R}^{d_{\text{SAE}}}$ denote the sparse feature vector for segment x. For each feature dimension i, we compute the difference in mean activation

$$\Delta_i = \mathbb{E}[f_i(x) \mid x \in X^+] - \mathbb{E}[f_i(x) \mid x \in X^-]. \quad (1)$$

Feature selection is done on the training split only.

The pipeline estimates document clustered permutation statistics and bootstrap selection frequencies, applies Benjamini Hochberg correction, and then selects the top positive feature set. This is crucial because layer choice, feature choice, and thresholding all count as model selection. Test data are never used to choose the layer or the feature bank.

4.7 Held Out Transfer Evaluation

Discovery alone only tells us that a proxy can be separated. To test whether the discovered structure is reusable, we transfer a fixed feature bank from one proxy to its paired concept on held out data. If a source bank discovered for proxy p_s contains feature indices I with weights w, the held out transfer score is

$$S_{p_s \to p_t}(x) = \sum_{i \in I} w_i f_i(x). \tag{2}$$

We then compare within pair transfer to cross pair transfer. This is the central test for shared policy feature structure. A reusable policy feature bank should perform better on its related target than on unrelated out of pair targets.

4.8 Keyword Masking Robustness

To test whether transfer is driven by shallow lexical triggers, we rerun evaluation after masking proxy specific keywords. Because the masked run reuses the exact same evaluation rows, the comparison isolates lexical stress rather than a change in dataset composition. This robustness check is essential for distinguishing genuinely useful internal policy features from sparse lexical shortcuts.

4.9 Family Core Feature Consolidation

When multiple related proxies repeatedly activate overlapping sparse features, we consolidate them into a family core bank. A feature enters a family core only if it appears in at least two related proxies and exceeds a minimum bootstrap stability threshold. This step separates one off proxy discoveries from repeated structure that may support stronger interpretation.

4.10 Causal Intervention on Held Out Policy Judgments

The final methodological step is intervention. We ablate or clamp selected sparse family core features during inference and observe how held out policy judgments change. The current implementation uses held out forced choice policy tasks and compares the target margin drop against matched random feature controls. This turns the representation question into a mechanistic one: not only are features discoverable and transferable, but some of them may be selectively involved in downstream policy judgments.

Proxy task	Test positives	Validated test positives
Bias	43	12
Discrimination	40	10
Privacy	28	11
Rights violation	51	18
Transparency	47	14
Interpretability	6	3

Table 1: Held out positive counts for the six core proxy tasks in the current policy representation study. These counts define the empirical scope of the v1 analysis and explain why later claims on the transparency and interpretability pair must remain cautious.

5 Experimental Setup

5.1 Policy Feature Governance Audit Benchmark

To make evaluation systematic, we package the methodology above into a compact audit protocol, the *Policy Feature Governance Audit Benchmark*. In the paper, this benchmark is not the main contribution. Its role is to standardize how we compare methods on the same policy representation tasks.

The benchmark compares four methods:

1. TF IDF logistic regression
2. sentence embedding logistic regression
3. dense residual logistic regression
4. sparse SAE feature banks

The benchmark contributes four axes:

1. **Coverage**, which measures held out proxy recovery
2. **Consistency**, which measures within pair versus out of pair transfer
3. **Robustness**, which measures retention under keyword masking
4. **Causality**, which measures selective behavioral effects under sparse feature intervention

Only the first three axes define the core method comparison. Causality is reported separately as a mechanistic qualification because only the sparse internal method supports direct intervention in the current setup.

5.2 Current Data Readiness

The current corpus build contains 1,965 training segments, 445 development segments, and 309 test segments after quality filtering. The validated subset contains 403, 106, and 52 segments respectively. Table 1 lists the held out positive counts for the six current proxy tasks.

Preflight check	Status
Fixed six task registry	Pass
Document leakage check	Pass
Reference split counts	Pass
Validated rerun availability	Pass
Matched negatives present	Pass

Table 2: Current preflight status for the policy feature benchmark on the real AGORA build.

5.3 Preflight Protocol

Before any benchmark numbers are trusted, the current implementation runs a preflight stage. Preflight verifies that the six fixed tasks are present, document leakage is absent, reference split counts match the manifests, validated reruns exist, and matched negatives are available for every task. The current real build passes all of these checks.

6 Results

At this stage, the paper reports implementation readiness and benchmark preflight results rather than final full model comparisons. This is appropriate because the expensive pretrained Gemma 2 2B plus Gemma Scope run has not yet been completed in this document.

6.1 Preflight and Readiness Results

The current system passes all benchmark preflight checks. The six task registry is intact, the document grouped split remains leakage free, current split counts match the stored reference counts, validated reruns are available, and matched negatives exist for all tasks. Table 2 summarizes the current preflight state.

6.2 What the Full Results Will Test

When the full run is executed, the key empirical sequence will be the following. First, we will measure coverage to determine which policy features are internally recoverable. Second, we will evaluate transfer across the three concept pairs to determine whether related policy features share reusable structure. Third, we will test robustness under keyword masking to estimate how much of that signal survives lexical stress. Finally, we will evaluate sparse family core interventions to determine whether the discovered internal features are causally involved in held out policy judgments.

7 Discussion

The main scientific risk in this project is over interpreting proxy specific structure as a broad normative or conceptual representation. A model may separate AGORA proxies using local wording, drafting conventions, or institutional artifacts without representing a broader policy concept. The methodology is designed to make that risk measurable rather than implicit.

A single proxy discovery run only supports a proxy aligned feature claim. Stronger evidence requires fixed held out evaluation, better within pair than out of pair transfer, resilience under keyword masking, and selective causal effects under intervention. If these conditions are met, then it becomes reasonable to describe the resulting feature sets as shared policy feature structure and, more cautiously, as structure that is consistent with higher level policy concept families.

This is also where the technical governance contribution sits. The benchmark is valuable not because it replaces the methodology, but because it turns the methodology into a repeatable audit protocol. If a method recovers policy features that are separable, transferable, robust, and causally relevant, then internal policy representation becomes something we can compare and monitor rather than only speculate about.

8 Limitations

This paper has several limitations.

First, the current document does not yet report the full expensive Gemma 2 2B pretrained plus Gemma Scope benchmark outputs. It therefore presents the completed methodology and evaluation protocol together with readiness evidence, but not the final substantive scores.

Second, some proxy tasks remain sparse. In particular, interpretability has only six held out positive examples in the current main test split and three in the validated test split. That is enough for inclusion in the fixed v1 suite, but it limits confidence in any later fine grained claim on that pair.

Third, matched negatives reduce shallow shortcuts but do not eliminate all structured dependence. Policy text remains clustered by institution, jurisdiction, and drafting style, and some discovered internal structure may still reflect those factors.

Fourth, public values remain an interpretive extension rather than a direct empirical target. The current paper does not claim that language models directly encode public values. It only builds a path for testing whether shared policy feature structure might later support that kind of interpretation.

Finally, automated circuit discovery remains outside the main narrative in this version. Circuit extraction may become useful later as an appendix or phase two analysis, but it is not needed for the current core contribution.

9 Conclusion

We reframed the study around a central mechanistic question: how do language models internally represent and process policy text? Using AGORA policy annotations as observable proxies, we developed a proxy grounded methodology that combines matched negatives, dense and sparse internal features, train only discovery, held out transfer, lexical stress testing, family core consolidation, and intervention based causal evaluation. We also packaged the evaluation component

into a compact audit benchmark, but kept that benchmark in a supporting role. The current implementation already supports the full pipeline over 2,719 filtered AGORA segments and passes all preflight checks. The immediate next step is to run the full pretrained Gemma Scope experiments and populate the now standardized coverage, transfer, robustness, and causal outputs.

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